

- Description

This module manages an automatic interest rate adjustment system for bank loans, ensuring that:

1. The system continuously monitors the central bank's reference interest rate (in percentage).
2. If the reference rate exceeds a predefined threshold, the loan interest rate is increased accordingly.
3. If the reference rate is below the threshold, the loan interest rate remains unchanged.

This system helps financial institutions adjust loan rates dynamically based on economic conditions.

- Specification

module BankLoanRateControl

var currentReferenceRate, referenceRateThreshold, baseInterestRate,
currentLoanRate: real

process updateReferenceRate(newReferenceRate: real) returns (currentLoanRate: real)

functional scenario form:

currentReferenceRate > referenceRateThreshold && currentLoanRate = baseInterestRate + (currentReferenceRate - referenceRateThreshold) * 0.5, output "ALERT: High reference rate (" + currentReferenceRate + "%). Loan interest rate adjusted to " + currentLoanRate + "%.", Loan interest rate increased.

||

currentReferenceRate <= referenceRateThreshold && currentLoanRate = baseInterestRate, output "Reference rate normal (" + currentReferenceRate + "%). Loan interest rate remains " + currentLoanRate + "%.", Loan interest rate remains unchanged.

end process

end module

- Java Code

```
package code;
public class c98_BankLoanSystem {
    private double baseInterestRate;
    private double referenceRateThreshold;
    private double currentReferenceRate;
    private double currentLoanRate;
    public c98_BankLoanSystem(double baseInterestRate, double reference
RateThreshold) {
        this.baseInterestRate = baseInterestRate;
        this.referenceRateThreshold = referenceRateThreshold;
        this.currentReferenceRate = 0.0;
        this.currentLoanRate = baseInterestRate;
    }
}
```

```

    }
    public void updateReferenceRate(double newReferenceRate) {
        currentReferenceRate = newReferenceRate;
        adjustLoanRate();
        System.out.printf("Current Reference Rate: %.2f%%, Loan Interest Rate: %.2f%%\n",
            currentReferenceRate, currentLoanRate);
        if (currentReferenceRate > referenceRateThreshold) {
            System.out.println("ALERT: High reference rate detected. Loan interest rate increased.");
        } else {
            System.out.println("Reference rate within limits. Loan interest rate unchanged.");
        }
    }
}

private void adjustLoanRate() {
    if (currentReferenceRate > referenceRateThreshold) {
        currentLoanRate = baseInterestRate + (currentReferenceRate - referenceRateThreshold) * 0.5;
    } else {
        currentLoanRate = baseInterestRate;
    }
}

public double getCurrentLoanRate() {
    return currentLoanRate;
}

public static void main(String[] args) {
    c98_BankLoanSystem bankLoan = new c98_BankLoanSystem(5.0, 3.0);
    // testcase-VT:
    //     bankLoan.updateReferenceRate(2.5);
    //     bankLoan.updateReferenceRate(3.5);
    //     bankLoan.updateReferenceRate(3.0);

    // testcase-FT:
    //     bankLoan.updateReferenceRate(-1.7836486571938841E308);
    //     bankLoan.updateReferenceRate(-1.1586598667603788E308);
    //     bankLoan.updateReferenceRate(-1.1235481369576206E308);

    // testcase-Z3:
    //     bankLoan.updateReferenceRate(2.5);
    //     bankLoan.updateReferenceRate(3.5);
    //     bankLoan.updateReferenceRate(3.0);

    // testcase-UVT:

```

```

        bankLoan.updateReferenceRate(2.5);
        bankLoan.updateReferenceRate(3.5);
        bankLoan.updateReferenceRate(3.0);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c98_BankLoanSystem.java	100.0 %	4	0	4
▼ c98_BankLoanSystem	100.0 %	4	0	4
● updateReferenceRate(double)	100.0 %	2	0	2
■ adjustLoanRate()	100.0 %	2	0	2
● c98_BankLoanSystem(double, d	100.0 %	0	0	0
● main(String[])	100.0 %	0	0	0

3. Vibration testing instruction coverage: 96.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	92	21,688	21,780
▼ src	0.4 %	92	21,688	21,780
▼ code	0.4 %	92	21,688	21,780
▼ c98_BankLoanSystem.java	96.8 %	92	3	95
▼ c98_BankLoanSystem	96.8 %	92	3	95
● updateReferenceRate(double)	100.0 %	37	0	37
■ adjustLoanRate()	100.0 %	24	0	24
● c98_BankLoanSystem(double, d	100.0 %	15	0	15
● main(String[])	100.0 %	16	0	16

4. Fuzz testing branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c98_BankLoanSystem.java	50.0 %	2	2	4
▼ c98_BankLoanSystem	50.0 %	2	2	4
● updateReferenceRate(double)	50.0 %	1	1	2
■ adjustLoanRate()	50.0 %	1	1	2
● c98_BankLoanSystem(double, d	100.0 %	0	0	0
● main(String[])	100.0 %	0	0	0

5. Fuzz testing instruction coverage: 78.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	75	21,705	21,780
▼ src	0.3 %	75	21,705	21,780
▼ code	0.3 %	75	21,705	21,780
▼ c98_BankLoanSystem.java	78.9 %	75	20	95
▼ c98_BankLoanSystem	78.9 %	75	20	95
● updateReferenceRate(double)	89.2 %	33	4	37
■ adjustLoanRate()	45.8 %	11	13	24
● c98_BankLoanSystem(double, d	100.0 %	15	0	15
● main(String[])	100.0 %	16	0	16

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c98_BankLoanSystem.java	100.0 %	4	0	4
▼ c98_BankLoanSystem	100.0 %	4	0	4
● updateReferenceRate(double)	100.0 %	2	0	2
■ adjustLoanRate()	100.0 %	2	0	2
● c98_BankLoanSystem(double, d		0	0	0
● main(String[])		0	0	0

7. Z3 instruction coverage: 96.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	92	21,688	21,780
▼ src	0.4 %	92	21,688	21,780
▼ code	0.4 %	92	21,688	21,780
▼ c98_BankLoanSystem.java	96.8 %	92	3	95
▼ c98_BankLoanSystem	96.8 %	92	3	95
● updateReferenceRate(double)	100.0 %	37	0	37
■ adjustLoanRate()	100.0 %	24	0	24
● c98_BankLoanSystem(double, d	100.0 %	15	0	15
● main(String[])	100.0 %	16	0	16

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c98_BankLoanSystem.java	100.0 %	4	0	4
▼ c98_BankLoanSystem	100.0 %	4	0	4
● updateReferenceRate(double)	100.0 %	2	0	2
■ adjustLoanRate()	100.0 %	2	0	2
● c98_BankLoanSystem(double, d		0	0	0
● main(String[])		0	0	0

9. Unimproved V-method instruction coverage: 96.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	92	21,688	21,780
▼ src	0.4 %	92	21,688	21,780
▼ code	0.4 %	92	21,688	21,780
▼ c98_BankLoanSystem.java	96.8 %	92	3	95
▼ c98_BankLoanSystem	96.8 %	92	3	95
● updateReferenceRate(double)	100.0 %	37	0	37
■ adjustLoanRate()	100.0 %	24	0	24
● c98_BankLoanSystem(double, d	100.0 %	15	0	15
● main(String[])	100.0 %	16	0	16